

【Overview】

Solar cell car "competition" than fuel cell car "competition". ver1.2.3

This is design plan of "the elemental technology patent".

燃料電池車競争よりも太陽電池（ソーラー）競争でしょう ver1.2.3

これは「要素技術特許」の構想案です。

【Industrial field of application】

産業上の利用分野

It is available in the wide field of industry, the industry of an energy-related field, the electricity-related field without being limited to the field of electronics.

エレクトロニクス分野に限られず、エネルギー関連分野、電力関連分野などの産業・工業の幅広い分野に利用できるものです。

【TITLE】

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燃料電池車競争よりも太陽電池（ソーラー）競争でしょう ver1.2.3

I abstracting my paper (content published on blog) that I previously "drafted".

以前に、私が「起稿」した論文の抄録（ブログで公開した内容）です。

【THEME 1】

Think about issues

Is not the limit of the fuel cell "already visible" ?

課題について考える

燃料電池の限界はもう見えている？

1. Internal resistance is too high, self-heating is too large.

It is the biggest disadvantage.

2. The inductive load such as motors has intense load change

fever in the low speed is relatively small,

place that a heat loss difference is bigger than a real electricity loss difference becoming extremely big, and the real heat radiation efficiency worsens from the thing that is in state that "it is imbalance", and "cooling means(device)" makes the fever in the high speed big and is hard to downsize is a difficult point

3. Conversion efficiency of the electricity is too bad.

Are the present conditions not 50% - around 60%?

High efficiency is pressed full-scale practical use for

If high efficiency is realized in a large electricity domain, self-fever is suppressed, and the quantity to occur of "the water" decreases.

Do you understand the advantage that the quantity that this "water" produces decreases?

4. The scientific battery (fuel cell) has high temperature dependence

When temperature falls, electromotive force tends to fall down,

When temperature rises; electromotive force the place in the buoyancy, however,

When temperature rises, according to "1 to 3", temperature due to the self-fever rises, and "cooling means(device)" makes is big "regulation (load curve properties)" badly after all

【THEME 2】

Think about issues

It will be solar cell competition

課題について考える

太陽電池（ソーラー）競争でしょう

1. It is a semiconductor device

As for the biggest problem of the solar, internal resistance is negative resistance (non-linear device) properties.

I do not turn to perform a direct "drive" of the inductive load such as motors.

2. As a method to solve a problem

In the place where high efficiency can be realized than the present fuel cell, I have utility more if I perform linearization of the internal resistance of the power supply through a regulator (stabilization power supply).

3. I depend on the device process

The practical use of the nanoproccess seems to be stable, but it will be a matter of time to become the "number nano" process from "dozens of nano" process.

The triple range of whom realization is pressed for semiconductors which became a three-dimensional device multi-layer solid from a two-dimensional device is not epitaxial, and the semiconductor of the high-dimensional multi-layer tertiary structure named watches of the night is necessary.

【Column】

An old tale

"VHS" and " β " (beta)

"DVD" and "BD Blu-ray"

"Liquid crystal" and "plasma"

May a similar choice come sometime soon?

When go for the energy saving era, inefficient high consumption electricity form "plasma" cannot spread in a general family.

See "the fool of the elite" of the high educational it.

Different viewpoints

Is it same as "a linear motor?" If high speed works, and "a semiconductor to watch in a dream" which can handle large electricity realizes it, "negative feedback control" (feedback) to control "back electromotive force" of the inductive load such as motors may develop the spread for making it more practically.

This is design plan of "the elemental technology patent"

Improvement of the motor

If there is such a "motor".

It is good if I can make "the magnet" which performs "absorption" of "back electromotive force" of the inductive load such as the motors which I listed earlier.

If absorption (restraint) is possible, it is over back electromotive force of the inductive load to suppress "the reverse withstand pressure" of the control circuit of the motor including the rectifying circuit low,

The order direction voltage can make an extremely small semiconductor if I can lower this reverse leak current (recovery electric current) and it is high-speed and, with the semiconductor which is smaller than "Schottky barrier diode", can control large electricity.

I can reduce the electrical "spikes noise" of the control system of the motor

Heat loss becomes small and can suppress the fever

The motor of the high efficiency can come out

The magnet which is indispensable in the next rare earth elements rare metal competition motor, the old days,

- Ferrite
- Cobalt
- Alnico1
- Amorphous
- Neodymium
- Is next ... something?

Where is a country getting the next "rare earth elements rare metal?"

[Reference material in summary]

From the Yomiuri Shimbun

Fuel cell-powered vehicle competition low price realization grasps the key to spread (the Yomiuri editorial dated July 17, 2013)

[comment]

The price reduction,

With balance with the demand that consumers demand, I think it to be the outcome of "the result of the necessity".

Fuel cell

Solar battery (solar)

Hybrid of lithium

Three fuse, and do you not become a hybrid newly?

I can apply for a utility model even immediately, but, from the reason that I spoke so far, have already seen the limit of the fuel cell.

The energy to continue in next-generation (the future) will be "solar battery".

...Thinking about

Dewa mata. see you.

【Created date】

Date created and released in Japanese sentences

日本文で作成し公開した日

2013 年 07 月 20 日（土）頃 Publish to my blog

2017 年 10 月 14 日（土） I show only the summary of the above sentence part of it Publish to your facebook

Publish to your facebook

Date translated and published in English

Created date : October 15, 2017 (Sun)

英文に翻訳して作成し公開した日 : 2017 年 10 月 15 日（日）

【Self-introduction】

Talk of music and sound of "nagagutsu o haita cello"

Murakami kōji 村上光治 （nagagutsu o haita cello） desu

Dewa mata. see you.

【About disclosure】

About disclosure On Saturday, July 20, 2013, we translate the contents made in Japanese into "blog" and publish it is a English.

公開について、2013 年 07 月 20 日（土）頃に、日本語で作成しブログ公開した内容を英語に翻訳して公開

【Author】

kouji murakami 村上光治

Talk of music and sound of "nagagutsu o haita cello"

Murakami kōji 村上光治 （nagagutsu o haita cello） desu

Dewa mata. see you.

【自己紹介】

国立国会図書館「<http://ndl.go.jp/>」→「村上光治」検索

科学技術総合リンクセンター「<http://jglobal.jst.go.jp/>」→「村上光治」検索 他にも多数有

【ファイル追加公開】

応用数学 Applied Mathematics

<http://kouji-murakami1.cocolog-nifty.com/>

<http://kouji-murakami1.cocolog-nifty.com/blog/2010/01/applied-mathe-1.html>

http://kouji-murakami1.cocolog-nifty.com/blog/2010/02circuit/MJ_awarded.pdf

【関連サイト】

音楽と音響の調和 （「放送技術」 連載「音楽と音響の調和」を PDF 再編成）

<http://kouji-murakami2.cocolog-nifty.com/>

<http://kouji-murakami2.cocolog-nifty.com/blog/2015/07/kouji-murakam-2.html>

【Public list PDF】

Turn to the next-generation Youtube challenge

次世代に向けた「YouTube」の課題

Public No : 001.5

http://kouji-murakami2.cocolog-nifty.com/blog/2017/Public/Public_No_001.5.pdf

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Public No : 102.3 #001

http://kouji-murakami2.cocolog-nifty.com/blog/2017/Public/Public_No_102.3.pdf

【Correction】

2017 年 10 月 20 日（金）#001 Address revisio

<https://www.facebook.com/nohcello>

<http://kouji-murakami1.cocolog-nifty.com/>

<http://kouji-murakami2.cocolog-nifty.com/>